GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper A

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

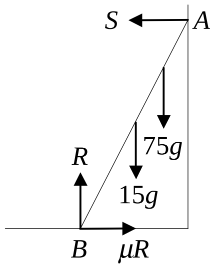
M2 Paper A – Marking Guide

1.	cons. of mom: $m(5) - m(3) = mv_1 + mv_2$		M1	
	$v_1 + v_2 = 2$		A1	
	$\frac{v_2 - v_1}{5-(-3)} = \frac{1}{2} \quad \therefore v_2 - v_1 = 4$		M1 A1	
	solve simul. giving $v_1 = {}^{-}1 \text{ ms}^{-1}$ so speed is 1 ms^{-1}, $v_2 = 3 \text{ ms}^{-1}$		M1 A1	**(6)**

2.	*(a)*	$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (2t - 3)\mathbf{i} + \frac{1}{2}t^2\mathbf{j}$	M1 A1	
		when $t = 0$, $\mathbf{v} = {}^{-}3\mathbf{i} \text{ ms}^{-1}$	A1	
	(b)	at $t = 2$, $\mathbf{v} = \mathbf{i} + 2\mathbf{j} \quad \therefore \lvert \mathbf{v} \rvert = \sqrt{(1^2 + 2^2)} = \sqrt{5}$	M2 A1	
		KE lost = $\frac{1}{2}(3)(3^2 - 5) = 6\text{J}$	M1 A1	**(8)**

3.	*(a)*	(i) uniform rod	B1	
		(ii) particle	B1	

(b)



resolve ↑:	$R - 15g - 75g = 0 \quad \therefore R = 90g$	M1	
resolve →:	$\mu R - S = 0 \quad \therefore S = 30g$	M1 A1	
mom. about B	$S.8\sin\theta - 15g.4\cos\theta - 75g.d\cos\theta = 0$	M1 A1	
	$8S\tan\theta - 60g = 75gd$	M1	
	$d = \frac{420g}{75g} = 5.6 \quad \therefore AP = 8 - 5.6 = 2.4 \text{ m}$	M1 A1	**(10)**

4.	*(a)*	$a \propto (3t^2 - 5) \quad \therefore \quad a = k(3t^2 - 5)$	M1	
		$v = \int a \, dt = k(t^3 - 5t) + c$	M1 A1	
		when $t = 0$, $v = 0$ so $c = 0$	A1	
		when $t = 3$, $v = 3$ so $3 = k(27 - 15) \quad \therefore \quad k = \frac{1}{4}$	M1 A1	
		$a = \frac{1}{4}(3t^2 - 5)$	A1	
	(b)	$s = \int v \, dt = \frac{1}{4}(\frac{1}{4}t^4 - \frac{5}{2}t^2) + c$	M1 A1	
		when $t = 0$, $s = 0$ so $c = 0 \quad \therefore \quad s = \frac{1}{4}(\frac{1}{4}t^4 - \frac{5}{2}t^2)$	M1	
		$s = \frac{1}{4}t^2(\frac{1}{4}t^2 - \frac{10}{4}) = \frac{1}{16}t^2(t^2 - 10)$	A1	**(11)**

---

5. *(a), (b)*

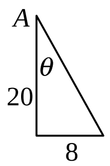
portion	mass	x	y	mx	my
ABFG	256ρ	4	16	1024ρ	4096ρ
CDEF	128ρ	16	4	2048ρ	512ρ
total	384ρ	$\bar{x}$	$\bar{y}$	3072ρ	4608ρ

ρ = mass per unit area x, y coords. taken horiz. / vert. from G — M3 A2

$\bar{x} = \frac{3072\rho}{384\rho} = 8$ so must lie on BF — M1 A1

$\bar{y} = \frac{4608\rho}{384\rho} = 12$ $\therefore$ dist. from AB = 20 cm — M1 A1

(c)



M1

$\tan\theta = \frac{8}{20}$ $\therefore$ $\theta = 21.8°$ (1dp) — M1 A1 **(12)**

---

6. *(a)* $\frac{P}{v} - R = ma$ $\therefore$ $\frac{90000}{20} - 1800 = 1200a$ — M2 A1

$\therefore$ $a = 2.25 \text{ ms}^{-2}$ — A1

(b) at max. speed, $a = 0$, $\frac{P}{v} - R = 0$ $\therefore$ $\frac{90000}{v} - 1800 = 0$ so $v = 50 \text{ ms}^{-1}$ — M1 A1

KE = $\frac{1}{2} \times 1200 \times 50^2 = 1\,500\,000$ J = 1500 kJ — M1 A1

(c) $\frac{P}{v} - R - mg\sin\alpha = 0$ $\therefore$ $\frac{90000}{25} - 1800 - 1200(9.8)\sin\alpha = 0$ — M2 A1

$\sin\alpha = \frac{1.5}{9.8}$ $\therefore$ $\alpha = 8.8°$ (1dp) — M1 A1 **(13)**

---

7. *(a)* particle moving freely under gravity — B2

(b) vert. disp. = 0 $\therefore t(u\sin\alpha - \frac{1}{2}gt) = 0$ — M1

$t = 0$ at O, we require $49\sin30° - 4.9t = 0$ $\therefore t = 5$ — M1 A1

horiz. disp. = $ut\cos\alpha = 49(5)\cos30° = 212.17$ — M1 A1

i.e. 212 – 170 beyond hole = 42.2 m (3sf) — A1

(c) when horiz. disp. = 170, $ut\cos\alpha = 170$ $\therefore$ $t = 4.006$ — M1

horiz. vel. = $u\cos\alpha = 42.44$ vert. vel. = $u\sin\alpha - gt = {}^{-}14.76$ — A2

mag. of vel = $\sqrt{[(42.44)^2 + ({}^{-}14.76)^2]} = 44.9 \text{ ms}^{-1}$ (3sf) — M1 A1

req'd angle = $\tan^{-1}\frac{14.76}{42.44}$ = 19.2° below horizontal (3sf) — M1 A1 **(15)**

Total **(75)**

Performance Record – M2 Paper A

Question no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	Total
Topic(s)	collisions	**i**, **j** calculus	statics ladder prob.	variable accel.	centre of mass	power, KE	projectiles	
Marks	6	8	10	11	12	13	15	75
Student								

GCE Examinations

Mechanics
Module M2

Advanced Subsidiary / Advanced Level

Paper A

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use $g = 9.8\ \text{m s}^{-2}$.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. Two identical particles are approaching each other along a straight horizontal track. Just before they collide, they are moving with speeds 5 m s^{-1} and 3 m s^{-1} respectively. The coefficient of restitution between the particles is $\frac{1}{2}$.

Find the speeds of the particles immediately after the impact. **(6 marks)**

---

2. A particle P of mass 3 kg moves such that at time t seconds its position vector, $\mathbf{r}$ metres, relative to a fixed origin, O, is given by

$$\mathbf{r} = (t^2 - 3t)\mathbf{i} + \tfrac{1}{6}t^3\mathbf{j},$$

where $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.

(a) Find the velocity of P when $t = 0$. **(3 marks)**

(b) Find the kinetic energy lost by P in the interval $0 \leq t \leq 2$. **(5 marks)**

3.

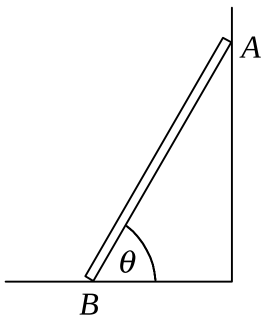


Fig. 1

Figure 1 shows a uniform ladder of mass 15 kg and length 8 m which rests against a smooth vertical wall at A with its lower end on rough horizontal ground at B. The coefficient of friction between the ladder and the ground is $\frac{1}{3}$ and the ladder is inclined at an angle θ to the horizontal, where $\tan\theta = 2$.

A man of mass 75 kg ascends the ladder until he reaches a point P. The ladder is then on the point of slipping.

(a) Write down suitable models for

(i) the ladder,

(ii) the man. **(2 marks)**

(b) Find the distance AP. **(8 marks)**

---

4. A particle P moves in a straight horizontal line such that its acceleration at time t seconds is proportional to $(3t^2 - 5)$.

Given that at time $t = 0$, P is at rest at the origin O and that at time $t = 3$, its velocity is 3 m s^{-1},

(a) find, in m s^{-2}, the acceleration of P in terms of t, **(7 marks)**

(b) show that the displacement of the particle, s metres, from O at time t is given by

$$s = \tfrac{1}{16}t^2(t^2 - 10)\,.$$ **(4 marks)**

5.

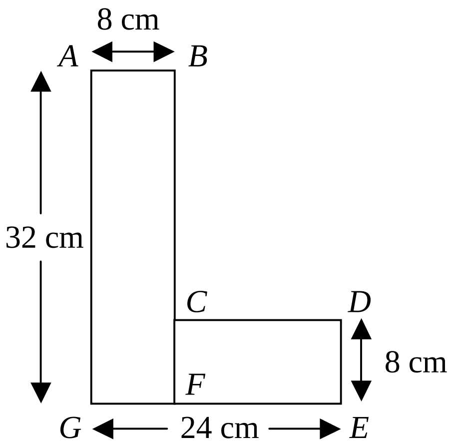


Fig. 2

Figure 2 shows a uniform plane lamina *ABCDEG* in the shape of a letter 'L' consisting of a rectangle *ABFG* joined to another rectangle *CDEF*. The sides *AB* and *DE* are both 8 cm long and the sides *EG* and *GA* are of length 24 cm and 32 cm respectively.

(a) Show that the centre of mass of the lamina lies on the line *BF*. **(5 marks)**

(b) Find the distance of the centre of mass from the line *AB*. **(4 marks)**

The uniform lamina in Figure 2 is a model of the letter 'L' in a sign above a shop. The letter is normally suspended from a wall at *A* and *B* so that *AB* is horizontal but the fixing at *B* has broken and the letter hangs in equilibrium from the point *A*.

(c) Find, in degrees to one decimal place, the acute angle *AG* makes with the vertical.

(3 marks)

Turn over

6. The engine of a car of mass 1200 kg is working at a constant rate of 90 kW as the car moves along a straight horizontal road. The resistive forces opposing the motion of the car are constant and of magnitude 1800 N.

(a) Find the acceleration of the car when it is travelling at 20 m s^{-1}. **(4 marks)**

(b) Find, in kJ, the kinetic energy of the car when it is travelling at maximum speed. **(4 marks)**

The car ascends a hill which is straight and makes an angle α with the horizontal. The power output of the engine and the non-gravitational forces opposing the motion remain the same.

Given that the car can attain a maximum speed of 25 m s^{-1},

(c) find, in degrees correct to one decimal place, the value of α. **(5 marks)**

7.

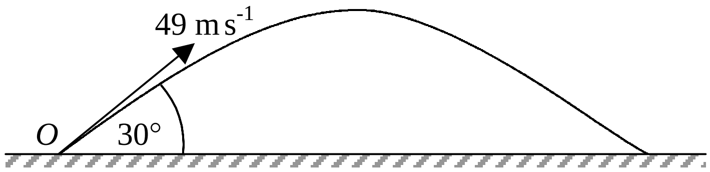


Fig. 3

Figure 3 shows the path of a golf ball which is hit from the point O with speed 49 m s^{-1} at an angle of 30° to the horizontal. The path of the ball is in a vertical plane containing O and the hole at which the ball is aimed.

The hole is 170 m from O and on the same horizontal level as O.

(a) Suggest a suitable model for the motion of the golf ball. **(2 marks)**

Find, correct to 3 significant figures,

(b) the distance beyond the hole at which the ball hits the ground, **(6 marks)**

(c) the magnitude and direction of the velocity of the ball when it is directly above the hole. **(7 marks)**

END

GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper B

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

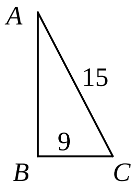
(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

M2 Paper B – Marking Guide

1.	*(a)*	work done = force $\times$ dist. = $8000 \times 0.04 = 320$ J	M1 A1	
	(b)	work done = change in KE = $\frac{1}{2}m(v^2 - u^2)$	M1	
		$= \frac{1}{2}0.025(v^2 - 200^2)$ $\therefore$ $v^2 - 40\,000 = {}^{-}25\,600$	M2 A1	
		$v^2 = 14\,400$ $\therefore$ $v = 120$ m s^{-1}	A1	**(7)**
2.	*(a)*	at max. speed, $a = 0$, $\frac{P}{v} - R = 0$ $\therefore$ $\frac{P}{30} - 2000 = 0$	M1 A1	
		$P = 60\,000$ W $\therefore$ $H = 60$	A1	
	(b)	$1.2 \times 60 = 72$	A1	
		$\frac{P}{v} - R = ma$ $\therefore$ $\frac{72000}{30} - 2000 = m \times 0.32$	M1 A1	
		$400 = 0.32m$ $\therefore$ $m = 1250$ kg	A1	**(7)**

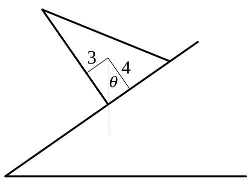
3. *(a)*



(i)	c.o.m. = $\frac{1}{3}$ dist. from B to C = $\frac{1}{3} \times 9 = 3$ cm from AB	M1 A1	
(ii)	$AB = \sqrt{(15^2 - 9^2)} = 12$ cm	M1 A1	
	c.o.m. = $\frac{1}{3}$ dist. from B to A = $\frac{1}{3} \times 12 = 4$ cm from BC	M1 A1	

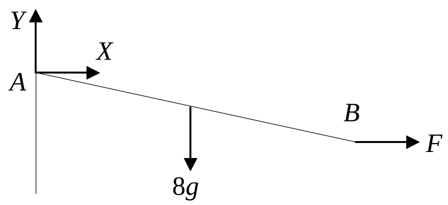
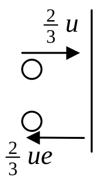
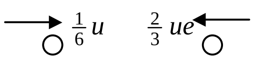
(b) lamina will not topple if vertical through c.o.m. passes between B and C — B1

max. θ when it passes through B — B1



$\tan\theta = \frac{3}{4}$ $\therefore$ $\theta = 36.9°$ (1dp) — M1 A1 **(10)**

4.	*(a)*	$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 3\mathbf{i} - 2t\mathbf{j}$ and when $t = 2$, $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$	M1 A1	
		mag. of $\mathbf{a} = \sqrt{[(3)^2 + ({}^{-}4)^2]} = 5$ m s^{-2}	M1 A1	
	(b)	$s = \int v\,dt = \frac{3}{2}t^2\mathbf{i} - \frac{1}{3}t^3\mathbf{j} + A\mathbf{i} + B\mathbf{j}$	M1 A1	
		when $t = 0$, $s = 6\mathbf{i} + 12\mathbf{j}$ so $A = 6$, $B = 12$	M1	
		$s = (\frac{3}{2}t^2 + 6)\mathbf{i} + (12 - \frac{1}{3}t^3)\mathbf{j}$	A1	
		disp. when $t = 6$ is $60\mathbf{i} - 60\mathbf{j} = 60(\mathbf{i} - \mathbf{j})$ $\therefore$ $k = 60$	M1 A1	**(10)**

5.	*(a)*			
		mom. about A $\quad 8g(a\cos20°) - F(2a\sin20°) = 0$	M1 A1	
		$F = \frac{4g}{\tan 20°} = 108$ N (3sf)	M1 A1	
	(b)	resolve $\rightarrow$: $\quad F + X = 0 \quad \therefore X = {}^{-}108$	A1	
		resolve $\uparrow$: $\quad Y - 8g = 0 \quad \therefore Y = 78.4$ N	A1	
		mag. of reaction at hinge $= \sqrt{[({}^{-}108)^2 + (78.4)^2]} = 133$ N (3sf)	M1 A1	
		req'd angle $= \tan^{-1}\frac{108}{78.4} = 54°$ (nearest degree) to the vertical	M1 A1	**(10)**
6.	*(a)*	$s_y = (ut\sin\alpha - \frac{1}{2}gt^2) \;= t(u\sin\alpha - \frac{1}{2}gt)$	M1 A1	
		$s_y = 0$ when $t = 0$ (at A) and when $t = \frac{2u}{g}\sin\alpha$ (at B)	A1	
		$s_x = ut\cos\alpha = u(\frac{2u}{g}\sin\alpha)\cos\alpha$ (at B)	M1	
		$= \frac{u^2}{g}(2\sin\alpha\cos\alpha) = \frac{u^2}{g}\sin2\alpha$	M1 A1	
	(b)	$\frac{u^2}{g}\sin2\alpha = 80 \quad \therefore \quad \frac{45^2}{9.8}\sin2\alpha = 80$	M1 A1	
		$\sin2\alpha = 0.387$ $\quad$ giving $\alpha = 11.4°, 78.6°$ (1dp)	M2 A1	
	(c)	11.4° as larger horiz. component of vel.	B2	
	(d)	$t = \frac{2\times45}{g}\sin(11.4°) = 1.8$ seconds (1dp)	M1 A1	**(15)**
7.	*(a)*	cons. of mom: $\;4m(u) + 0 = 4mv_1 + 5mv_2$	M1	
		$4u = 4v_1 + 5v_2$	A1	
		$\frac{v_2 - v_1}{u - 0} = \frac{1}{2} \quad \therefore \quad u = 2v_2 - 2v_1$	M1 A1	
		solve sim. eqns. to get $v_1 = \frac{1}{6}u, \; v_2 = \frac{2}{3}u$	M1 A1	
		$\therefore \; v_2 = \frac{4}{6}u = 4 \times v_1$	A1	
	(b)	 speed of B after collision with wall $= \frac{2}{3}ue$	M1 A1	
				
		cons. of mom: $\;4m(\frac{1}{6}u) - 5m(\frac{2}{3}ue) = 4mw_1 + 0$	M1 A1	
		$\frac{2}{3}u - \frac{10}{3}ue = 4w_1 \quad \therefore \; 12w_1 = 2u - 10ue$	A1	
		$\frac{0 - w_1}{\frac{1}{6}u + \frac{2}{3}ue} = \frac{1}{2} \quad \therefore \; {}^{-}w_1 = \frac{1}{12}u + \frac{1}{3}ue$ $\quad$ giving $\quad {}^{-}12w_1 = u + 4ue$	M1 A1	
		eliminating w_1 gives $\; u + 4ue + 2u - 10ue = 0$	M1	
		$3u = 6ue \quad \therefore \; e = \frac{1}{2}$	A1	**(16)**
			Total	**(75)**

Performance Record – M2 Paper B

Question no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	Total
Topic(s)	work - energy	power	centre of mass, toppling	**i**, **j** calculus	statics	projectiles	collisions	
Marks	7	7	10	10	10	15	16	75
Student								

GCE Examinations

Mechanics
Module M2

Advanced Subsidiary / Advanced Level

Paper B

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use $g = 9.8 \text{ m s}^{-2}$.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. A bullet of mass 25 g is fired directly at a fixed wooden block of thickness 4 cm and passes through it. When the bullet hits the block, it is travelling horizontally at 200 m s^{-1}. The block exerts a constant resistive force of 8000 N on the bullet.

(a) Find the work done by the block on the bullet. **(2 marks)**

By using the Work-Energy principle,

(b) show that the bullet emerges from the block with speed 120 m s^{-1}. **(5 marks)**

---

2. A car is travelling along a straight horizontal road against resistances to motion which are constant and total 2000 N. When the engine of the car is working at a rate of H kilowatts, the maximum speed of the car is 30 m s^{-1}.

(a) Find the value of H. **(3 marks)**

The car driver wishes to overtake another vehicle so she increases the rate of working of the engine by 20% and this results in an initial acceleration of 0.32 m s^{-2}. Assuming that the resistances to motion remain constant,

(b) find the mass of the car. **(4 marks)**

3.

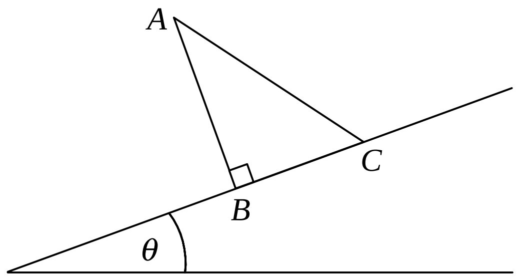


Fig. 1

Figure 1 shows a uniform triangular lamina ABC placed with edge BC along the line of greatest slope of a plane inclined at an angle θ to the horizontal. The lengths AC and BC are 15 cm and 9 cm respectively and $\angle ABC$ is a right angle.

(a) Find the distance of the centre of mass of the lamina from

(i) AB,

(ii) BC. **(6 marks)**

Assuming that the plane is rough enough to prevent the lamina from slipping,

(b) find in degrees, correct to 1 decimal place, the maximum value of θ for which the lamina remains in equilibrium. **(4 marks)**

---

4. The velocity $\mathbf{v}$ m s^{-1} of a particle P at time t seconds is given by $\mathbf{v} = 3t\mathbf{i} - t^2\mathbf{j}$.

(a) Find the magnitude of the acceleration of P when $t = 2$. **(4 marks)**

When $t = 0$, the displacement of P from a fixed origin O is $(6\mathbf{i} + 12\mathbf{j})$ m s^{-1}, where $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.

(b) Show that the displacement of P from O when $t = 6$ is given by $k(\mathbf{i} - \mathbf{j})$ m, where k is an integer which you should find.

(6 marks)

5.

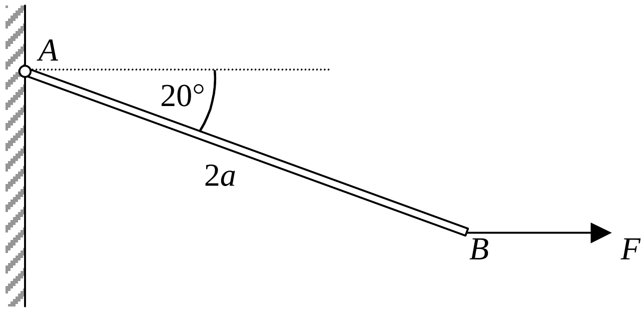


Fig. 2

A uniform rod AB of length $2a$ and mass 8 kg is smoothly hinged to a vertical wall at A.

The rod is held in equilibrium inclined at an angle of 20° to the horizontal by a force of magnitude F newtons acting horizontally at B which is below the level of A as shown in Figure 2.

(a) Find, correct to 3 significant figures, the value of F. **(4 marks)**

(b) Show that the magnitude of the reaction at the hinge is 133 N, correct to 3 significant figures, and find to the nearest degree the acute angle which the reaction makes with the vertical.

(6 marks)

Turn over

6. A particle P is projected from a point A on horizontal ground with speed u at an angle of elevation α and moves freely under gravity. P hits the ground at the point B.

(a) Show that $AB = \frac{u^2}{g}\sin 2\alpha$. **(6 marks)**

An archer fires an arrow with an initial speed of 45 m s^{-1} at a target which is level with the point of projection and at a distance of 80 m.

Given that the arrow hits the target,

(b) find in degrees, correct to 1 decimal place, the two possible angles of projection.

(5 marks)

(c) Write down, with a reason, which of the two possible angles of projection would give the shortest time of flight.

(2 marks)

(d) Show that the minimum time of flight is 1.8 seconds, correct to 1 decimal place.

(2 marks)

---

7. A smooth sphere A of mass $4m$ is moving on a smooth horizontal plane with speed u. It collides directly with a stationary smooth sphere B of mass $5m$ and with the same radius as A.

The coefficient of restitution between A and B is $\frac{1}{2}$.

(a) Show that after the collision the speed of B is 4 times greater than the speed of A.

(7 marks)

Sphere B subsequently hits a smooth vertical wall at right angles. After rebounding from the wall, B collides with A again and as a result of this collision, B comes to rest.

Given that the coefficient of restitution between B and the wall is e,

(b) find e. **(9 marks)**

END

GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper C

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

M2 Paper C – Marking Guide

1. *(a)* $\mathbf{v} = \frac{d\mathbf{r}}{dt} = 6t\mathbf{i} - 8t\mathbf{j}$ — M1 A1

$\mathbf{a} = \frac{d\mathbf{v}}{dt} = 6\mathbf{i} - 8\mathbf{j}$ not dependent on t so constant — M1 A1

(b) $\mathbf{F} = m\mathbf{a} = 2\mathbf{a} = 12\mathbf{i} - 16\mathbf{j}$ — A1

mag. of $\mathbf{F} = \sqrt{[(12)^2 + (^-16)^2]} = 20$ N — M1 A1 **(7)**

---

2. *(a)* X-sect. area of pipe $= \pi r^2 = \pi(0.05)^2$ — M1 A1

mass of water per second $= 6 \times 0.0025\pi \times 1000 = 15\pi$ — M1 A1

(b) energy gained $= \frac{1}{2}mv^2 + mgh = \frac{15}{2}\pi(6)^2 + (150\pi \times 9.8 \times 12)$ — M2 A1

$= 6390$ J $= 6.39$ kJ (3sf) — A1 **(8)**

---

3. *(a)* when $t = 0$, $v = 4$ ms^{-1} — A1

(b) particle at rest when $2t^2 - 9t + 4 = 0$ i.e. $(2t - 1)(t - 4) = 0$ — M1 A1

$t = \frac{1}{2}, 4$ — A1

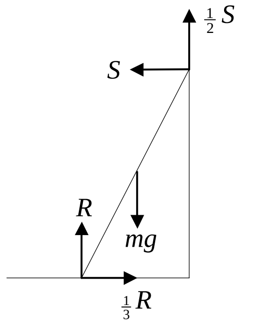
(c) $s = \int v\,dt = \frac{2}{3}t^3 - \frac{9}{2}t^2 + 4t + c$ — M1 A1

when $t = 0$, $s = 9$ so $c = 9$ $\therefore s = \frac{2}{3}t^3 - \frac{9}{2}t^2 + 4t + 9$ — A1

disp. when $t = 6$ is $\frac{2}{3}(6)^3 - \frac{9}{2}(6)^2 + 4(6) + 9$ — M1

$= 144 - 162 + 24 + 9 = 15$ m — A1 **(9)**

4.



resolve $\uparrow$: $\frac{1}{2}S + R - mg = 0$ — M1

resolve $\rightarrow$: $\frac{1}{3}R - S = 0$ — M1

solve simul. giving $S = \frac{1}{3}R$ $\therefore R = \frac{6}{7}mg$ — M1 A1

mom. about top of ladder $R.2a\cos\theta - \frac{1}{3}R.2a\sin\theta - mg.a\cos\theta = 0$ — M1 A1

$\therefore \tan\theta = \frac{2R - mg}{\frac{2}{3}R} = \frac{5}{4}$ — M2 A1 **(9)**

---

5. *(a)* vert. disp. = 0 $\therefore\ 8u_y - \frac{1}{2}g(8)^2 = 0$ — M1 A1

$u_y = \frac{1}{2}g(8) = 4g$ — A1

horiz. disp. = 24 $\therefore\ 8u_x = 24$ so $u_x = 3$ — M1 A1

(b) initial speed = $\sqrt{[(4g)^2 + 3^2]} = 39.3 \text{ ms}^{-1}$ (3sf) — M1 A1

(c) max. ht. when vert. vel = 0 $\therefore\ 0 = (4g)^2 - 2gs$ — M1 A1

$\therefore$ max. ht. = $8g$ = 78.4 m — A1

(d) e.g. small X-section, reasonable to treat as particle and ignore air res.
but, significant loss of mass during flight $\therefore$ model not very suitable — B3 **(13)**

---

6. *(a)* cons. of mom: $3mu + 0 = 3mv_1 + 2mv_2$ — M1

$\therefore\ 3v_1 + 2v_2 = 3u$ — A1

$\frac{v_2 - v_1}{u} = \frac{2}{3}$ $\therefore\ 3v_2 - 3v_1 = 2u$ — M1 A1

solve simul. giving $v_1 = \frac{1}{3}u$ and $v_2 = u$ — M1 A1

(b) cons. of mom: $2mu + 0 = 2mw_1 + 2mw_2$ — M1

$w_1 + w_2 = u$ — A1

$\frac{w_2 - w_1}{u} = e$ $\therefore\ w_2 - w_1 = eu$ — A1

solve simul. giving $w_1 = \frac{1}{2}u(1-e)$ — M1 A1

A and B collide again so speed of B < speed of A — M1

$\frac{1}{2}u(1-e) < \frac{1}{3}u$ so $\frac{1}{2}e > \frac{1}{2} - \frac{1}{3}$ $\therefore\ e > \frac{1}{3}$ — M1 A1 **(14)**

---

7. *(a)* from triangle properties, area of $BCD = \frac{1}{3}$ area of ABD — B1

$\therefore$ area of $BCD = \frac{1}{3}(\frac{1}{2} \times 2d \times \sqrt{3}d) = \frac{1}{3}\sqrt{3}d^2$ — M1 A1

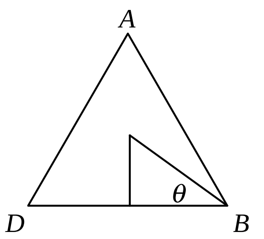
(b)

portion	mass	y	my
ABD	$\sqrt{3}d^2\rho$	$\frac{1}{3}\sqrt{3}d$	$d^3\rho$
BCD	$\frac{1}{3}\sqrt{3}d^2\rho$	$\frac{1}{9}\sqrt{3}d$	$\frac{1}{9}d^3\rho$
$ABCD$	$\frac{2}{3}\sqrt{3}d^2\rho$	$\bar{y}$	$\frac{8}{9}d^3\rho$

ρ = mass per unit area y coords. taken vert. from BD — M3 A3

$\bar{y} = \frac{\frac{8}{9}d^3\rho}{\frac{2}{3}\sqrt{3}d^2\rho} = \frac{4d}{3\sqrt{3}} = \frac{4}{9}\sqrt{3}d$ — M1 A1

(c)



$\theta = \tan^{-1}\frac{\frac{4}{9}\sqrt{3}d}{d} = \tan^{-1}\frac{4\sqrt{3}}{9}$ — M1 A1

req'd angle = $60 - \theta = 22.4°$ (1dp) — M1 A1 **(15)**

Total **(75)**

Performance Record – M2 Paper C

Question no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	Total
Topic(s)	**i**, **j** calculus	KE + PE	variable accel.	statics ladder prob.	projectiles	collisions	centre of mass	
Marks	7	8	9	9	13	14	15	75
Student								

GCE Examinations

Mechanics
Module M2

Advanced Subsidiary / Advanced Level

Paper C

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use g = 9.8 m s^{-2}.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. A particle P of mass 2 kg is subjected to a force $\mathbf{F}$ such that its displacement, $\mathbf{r}$ metres, from a fixed origin, O, at time t seconds is given by

$$\mathbf{r} = (3t^2 - 4)\mathbf{i} + (3 - 4t^2)\mathbf{j}.$$

(a) Show that the acceleration of P is constant. **(4 marks)**

(b) Find the magnitude of $\mathbf{F}$. **(3 marks)**

---

2. A pump raises water from a well 12 metres below the ground and ejects the water through a pipe of diameter 10 cm at a speed of 6 m s^{-1}.

Given that the mass of 1 m^3 of water is 1000 kg,

(a) find, in terms of π, the mass of water discharged by the pipe every second, **(4 marks)**

(b) find in kJ, correct to 3 significant figures, the total mechanical energy gained by the water per second.

(4 marks)

---

3. A particle moves in a straight horizontal line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 2t^2 - 9t + 4$. Initially, the particle has displacement 9 m from a fixed point O on the line.

(a) Find the initial velocity of the particle. **(1 mark)**

(b) Show that the particle is at rest when $t = 4$ and find the other value of t when it is at rest.

(3 marks)

(c) Find the displacement of the particle from O when $t = 6$. **(5 marks)**

4.

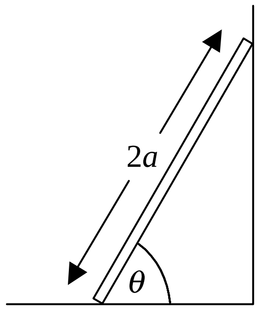


Fig. 1

Figure 1 shows a uniform ladder of mass m and length $2a$ resting against a rough vertical wall with its lower end on rough horizontal ground. The coefficient of friction between the ladder and the wall is $\frac{1}{2}$ and the coefficient of friction between the ladder and the ground is $\frac{1}{3}$.

Given that the ladder is in limiting equilibrium when it is inclined at an angle θ to the horizontal, show that $\tan\theta = \frac{5}{4}$.

(9 marks)

---

5. A firework company is testing its new brand of firework, the *Sputnik Special*. One of the company's employees lights a *Sputnik Special* on a large area of horizontal ground and it takes off at a small angle to the vertical. After a flight lasting 8 seconds it lands at a distance of 24 metres from the point where it was launched.

The employee models the firework as a particle and ignores air resistance and any loss of mass which the *Sputnik Special* experiences.

Using this model, find for this flight of the *Sputnik Special*,

(a) the horizontal and vertical components of the initial velocity, **(5 marks)**

(b) the initial speed, correct to 3 significant figures, **(2 marks)**

(c) the maximum height attained. **(3 marks)**

(d) Comment on the suitability of the modelling assumptions made by the employee.

(3 marks)

Turn over

6. Three uniform spheres A, B and C of equal radius have masses $3m$, $2m$ and $2m$ respectively. Initially, the spheres are at rest on a smooth horizontal table with their centres in a straight line and with B between A and C. Sphere A is projected directly towards B with speed u.

Given that the coefficient of restitution between A and B is $\frac{2}{3}$,

(a) show that the speeds of A and B after the collision are $\frac{1}{3}u$ and u respectively.

(6 marks)

The coefficient of restitution between B and C is e. Given that A and B collide again,

(b) show that $e > \frac{1}{3}$. **(8 marks)**

7.

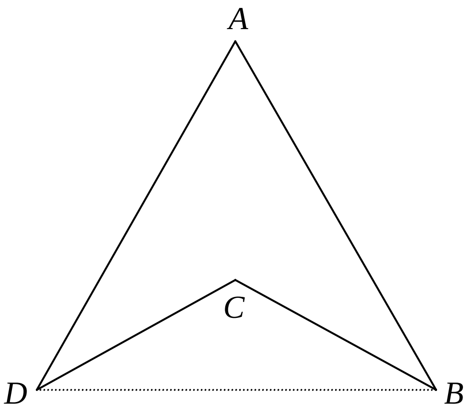


Fig. 2

Figure 2 shows a uniform lamina $ABCD$ formed by removing an isosceles triangle BCD from an equilateral triangle ABD of side $2d$. The point C is the centroid of triangle ABD.

(a) Find the area of triangle BCD in terms of d. **(3 marks)**

(b) Show that the distance of the centre of mass of the lamina from BD is $\frac{4}{9}\sqrt{3}\,d$.

(8 marks)

The lamina is freely suspended from the point B and hangs at rest.

(c) Find in degrees, correct to 1 decimal place, the acute angle that the side AB makes with the vertical.

(4 marks)

END

GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper D

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

M2 Paper D – Marking Guide

1. *(a)* $\mathbf{v} = \frac{d\mathbf{r}}{dt} = (3t - 3)\mathbf{i} + (t^2 - k)\mathbf{j}$ — M2 A1

(b) at rest when coeffs of **i** and **j** are both zero — M1

$3t - 3 = 0 \qquad t^2 - k = 0$ — M1

both satisfied when $k = 1$ — A1 **(6)**

---

2. cons. of mom: $2mu_1 - 5mu_2 = {}^-2m(3) + 5m(4)$

$2u_1 - 5u_2 = 14$ — M1 A1

$\frac{4-(-3)}{u_1+u_2} = \frac{1}{2} \quad \therefore u_1 + u_2 = 14$ — M1 A1

solve simul. giving $u_1 = 12 \text{ ms}^{-1}$, $u_2 = 2 \text{ ms}^{-1}$ — M1 A1 **(6)**

---

3. *(a)* $R \propto v \quad \therefore R = kv$, where k is a constant — M1

$\frac{P}{v} - R = 0 \quad \therefore \frac{90000}{50} - 50k = 0$ — M1 A1

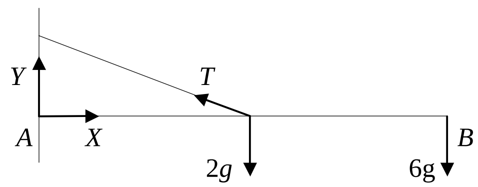
$k = 36 \quad \therefore R = 36v$ — A1

(b) $\frac{P}{v} - R - mg\sin\theta = 0 \quad \therefore \frac{90000}{v} - 36v - 1200(9.8)\frac{1}{14} = 0$ — M1 A1

$90000 - 36v^2 - 840v = 0 \quad \therefore 3v^2 + 70v - 7500 = 0$ — M1 A1

quad. form. giving $v = 39.7 \text{ ms}^{-1}$ (3sf) (clearly ${}^-63.0$ not suitable) — M1 A1 **(10)**

4.



(a) mom. about A $\quad 2ga + 6g(2a) - T\cos 60° = 0$ — M1 A1

$14ga = \frac{1}{2}Ta \quad \therefore T = 28g$ — M1 A1

(b) resolve $\uparrow$: $\quad Y + T\cos 60° - 8g = 0 \quad \therefore Y = {}^-6g$ — M1 A1

resolve $\rightarrow$: $\quad X - T\sin 60° = 0 \quad \therefore X = 14\sqrt{3}\,g$ — M1 A1

mag. of force at hinge = $\sqrt{[(14\sqrt{3}\,g)^2 + ({}^-6g)^2]} = 245$ N (3sf) — M1 A1

req'd angle = $\tan^{-1}\frac{6g}{14\sqrt{3}g} = 13.9°$ (3sf) below horizontal (away from wall) — M1 A1 **(12)**

---

5. *(a)* $v = \int a \, dt = 3t^2 - 10t + c$ — M1 A1

when $t = 0$, $v = 3$ so $c = 3 \quad \therefore v = 3t^2 - 10t + 3$ — M1 A1

$v = 0$ when $(3t - 1)(t - 3) = 0 \quad \therefore t = \frac{1}{3}, 3$ — M1 A1

(b) $s = \int v \, dt = t^3 - 5t^2 + 3t + k$ — M1 A1

when $t = 0$, $s = 0$ so $k = 0 \quad \therefore s = t^3 - 5t^2 + 3t$ — A1

disp. when $t = \frac{1}{3}$ is $(\frac{1}{3})^3 - 5(\frac{1}{3})^2 + 3(\frac{1}{3}) = \frac{13}{27}$ — M1 A1

disp. when $t = 2$ is $(2)^3 - 5(2)^2 + 3(2) = {}^-6$ — A1

dist. travelled $= 2 \times \frac{13}{27} + 6 = 6\frac{26}{27}$ m — A1 **(13)**

---

6. *(a)* min. α when ball passes through (12, $^{-}$0.6)

$12 = 14t\cos\alpha \quad \therefore\ t = \frac{6}{7\cos\alpha}$	M1 A1
$^{-}0.6 = 14t\sin\alpha - 4.9t^2$	M1
sub. in t giving $^{-}0.6 = 14(\frac{6}{7\cos\alpha})\sin\alpha - 4.9(\frac{6}{7\cos\alpha})^2$	A1
$^{-}0.6 = 12\tan\alpha - 3.6\sec^2\alpha$	M1
use $\sec^2\alpha \equiv 1 + \tan^2\alpha$ giving $6\tan^2\alpha - 20\tan\alpha + 5 = 0$	A1
use of quad. form. giving $\tan\alpha = 0.27$ (and 3.06)	
min. $\alpha = 15°$ (nearest degree)	M1 A1

(b)

$ut\cos\alpha = 12$		
$12 = 14t(\frac{3}{5}) \quad \therefore\ t = \frac{10}{7}$	M1 A1	
vert. disp., $ut\sin\alpha - \frac{1}{2}gt^2 = 14(\frac{10}{7})(\frac{4}{5}) - 4.9(\frac{10}{7})^2$	M1	
$= 16 - 10 = 6$	M1 A1	
i.e. 6 + 0.6 above $M \quad \therefore\ 6.6 - 2.4 = 4.2$m above crossbar	A1	**(14)**

---

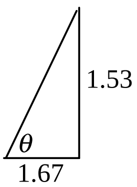
7. *(a)* (i), (ii)

portion	mass	x	y	mx	my
rectangle	32ρ	4	2	128ρ	64ρ
semicircle	$2\pi\rho$	6	$4 + \frac{8}{3\pi}$	$12\pi\rho$	$(8\pi + \frac{16}{3})\rho$
total	$(32 + 2\pi)\rho$	$\bar{x}$	$\bar{y}$	$(128 + 12\pi)\rho$	$(8\pi + \frac{208}{3})\rho$

ρ = mass per unit area $\quad x, y$ coords. taken horiz. / vert. from O	M4 A2
$\bar{x} = \frac{(128+12\pi)\rho}{(32+2\pi)\rho} = 4.33$ cm from OD (3sf)	M1 A1
$\bar{y} = \frac{(8\pi+\frac{208}{3})\rho}{(32+2\pi)\rho} = 2.47$ cm from OA (3sf)	M1A1

(b)

4 – 2.47 = 1.53 from m'pt. of BC vertically	M1
6 – 4.33 = 1.67 from m'pt. of BC horizontally	M1



$\tan\theta = \frac{1.53}{1.67} \quad \therefore\ \theta = 42.5°$ (3sf)	M1 A1	**(14)**

Total **(75)**

Performance Record – M2 Paper D

Question no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	Total
Topic(s)	**i**, **j** calculus	collisions	power	statics	variable accel.	projectiles	centre of mass	
Marks	6	6	10	12	13	14	14	75
Student								

GCE Examinations

Mechanics
Module M2

Advanced Subsidiary / Advanced Level

Paper D

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use $g = 9.8\ \text{m s}^{-2}$.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. A particle P moves such that at time t seconds its position vector, $\mathbf{r}$ metres, relative to a fixed origin O is given by

$$\mathbf{r} = (\tfrac{3}{2}t^2 - 3t)\mathbf{i} + (\tfrac{1}{3}t^3 - kt)\mathbf{j},$$

where k is a constant and $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors.

(a) Find an expression for the velocity of P at time t. **(3 marks)**

(b) Given that P comes to rest instantaneously, find the value of k. **(3 marks)**

---

2. Two smooth spheres P and Q of equal radius and of mass $2m$ and $5m$ respectively, are moving towards each other along a horizontal straight line when they collide. After the collision, P and Q travel in opposite directions with speeds of 3 m s^{-1} and 4 m s^{-1} respectively.

Given that the coefficient of restitution between the two particles is $\tfrac{1}{2}$, find the speeds of P and Q before the collision.

(6 marks)

---

3. A car of mass 1200 kg experiences a resistance to motion, R newtons, which is proportional to its speed, $v \text{ m s}^{-1}$. When the power output of the car engine is 90 kW and the car is travelling along a horizontal road, its maximum speed is 50 m s^{-1}.

(a) Show that $R = 36v$. **(4 marks)**

The car ascends a hill inclined at an angle θ to the horizontal where $\sin\theta = \tfrac{1}{14}$.

(b) Find, correct to 3 significant figures, the maximum speed of the car up the hill assuming that the power output of the engine is unchanged.

(6 marks)

4.

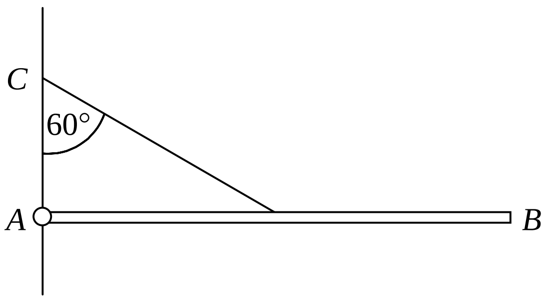


Fig. 1

Figure 1 shows a uniform rod *AB* of mass 2 kg and length 2*a*. The end *A* is attached by a smooth hinge to a fixed point on a vertical wall so that the rod can rotate freely in a vertical plane. A mass of 6 kg is placed at *B* and the rod is held in a horizontal position by a light string joining the midpoint of the rod to a point *C* on the wall, vertically above *A*. The string is inclined at an angle of 60° to the wall.

(a) Show that the tension in the string is 28*g*. **(4 marks)**

(b) Find the magnitude and direction of the force exerted by the hinge on the rod, giving your answers correct to 3 significant figures. **(8 marks)**

---

5. A particle *P* moves in a straight line with an acceleration of $(6t - 10)$ m s^{-2} at time *t* seconds. Initially *P* is at *O*, a fixed point on the line, and has velocity 3 m s^{-1}.

(a) Find the values of *t* for which the velocity of *P* is zero. **(6 marks)**

(b) Show that, during the first two seconds, *P* travels a distance of $6\frac{26}{27}$ m. **(7 marks)**

Turn over

6.

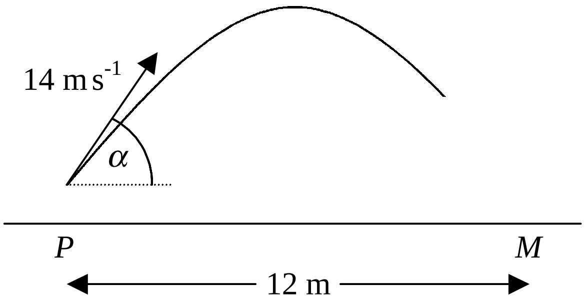


Fig. 2

A football player strikes a ball giving it an initial speed of 14 m s^{-1} at an angle α to the horizontal as shown in Figure 2. At the instant he strikes the ball it is 0.6 m vertically above the point *P* on the ground. The trajectory of the ball is in a vertical plane containing *P* and *M*, the middle of the goal-line. The distance between *P* and *M* is 12 m and the ground is horizontal.

Given that the ball passes over the point *M* without bouncing,

(a) find, to the nearest degree, the minimum value of α. **(8 marks)**

Given that the crossbar of the goal is 2.4 m above *M* and that $\tan \alpha = \frac{4}{3}$,

(b) show that the ball passes 4.2 m vertically above the crossbar. **(6 marks)**

7.

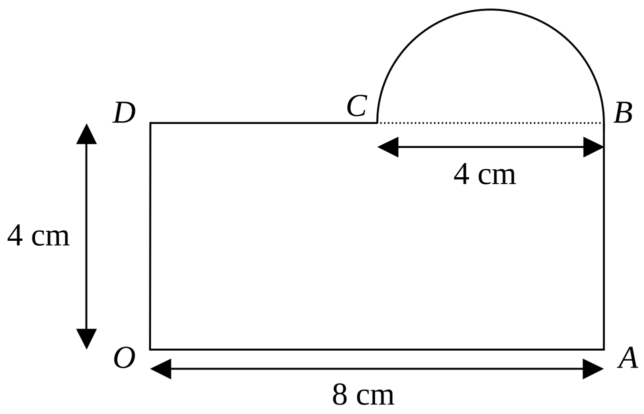


Fig. 3

Figure 3 shows a hotel 'key' consisting of a rectangle *OABD*, where *OA* = 8 cm and *OD* = 4 cm, joined to a semicircle whose diameter *BC* is 4 cm long. The thickness of the key is negligible and the same material is used throughout.

The key is modelled as a uniform lamina.

Using this model,

(a) find, correct to 3 significant figures, the distance of the centre of mass from

(i) *OD*,

(ii) *OA*. **(10 marks)**

A small circular hole of negligible diameter is made at the mid-point of *BC* so that the key can be hung on a smooth peg. When the key is freely suspended from the peg,

(b) find, correct to 3 significant figures, the acute angle made by *OA* with the vertical.

(4 marks)

END

GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper E

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

M2 Paper E – Marking Guide

1.	$\mathbf{I} = \Delta$mom. $\quad 12\mathbf{i} - 9\mathbf{j} = 0.6[(5\mathbf{i} + 3\mathbf{j}) - \mathbf{u}]$	M1 A1	
	$20\mathbf{i} - 15\mathbf{j} = 5\mathbf{i} + 3\mathbf{j} - \mathbf{u}$	M1	
	$\mathbf{u} = {}^{-}15\mathbf{i} + 18\mathbf{j}$	A1	**(4)**

---

2. *(a)* when $t = 0$, $x = 2 + 0 - \frac{1}{10} = 1.9$ m — M1 A1

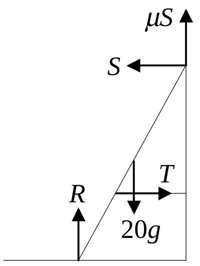
(b) $v = \frac{\mathrm{d}x}{\mathrm{d}t} = 1 - \frac{1}{10}\mathrm{e}^t$ — A1

at rest when $v = 0 \quad 1 - \frac{1}{10}\mathrm{e}^t = 0 \quad \therefore \quad \mathrm{e}^t = 10$ — M1 A1

$t = \ln 10 = 2.3$ (1dp) — A1 **(6)**

---

3. *(a)*



B2

(b) resolve $\uparrow$: $\quad R + \mu S - 20g = 0 \;\therefore\; R = 20g - \mu S$ — M1

resolve $\rightarrow$: $\quad T - S = 0 \;\therefore\; S = T$ — M1

eliminating S gives $\quad R = 20g - \frac{1}{3}T$ — A1

mom. about top of ladder $\quad T(4\sin\theta) + 20g(3\cos\theta) - R(6\cos\theta) = 0$ — M1 A1

$4T\tan\theta + 60g - 6R = 0$ — M1

$10T + 60g - 120g + 2T = 0 \;\therefore\; 12T = 60g$ and $T = 5g$ — A1

(c) attach rope lower down ladder/wall — B1

gives larger moment about top of ladder with same tension — B1 **(11)**

---

4. *(a)* (i), (ii)

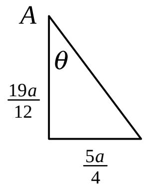
portion	mass	x	y	mx	my
AB	$2a\rho$	0	a	0	$2a^2\rho$
BC	$3a\rho$	$\frac{3}{2}a$	0	$\frac{9}{2}a^2\rho$	0
CD	$a\rho$	$3a$	$\frac{1}{2}a$	$3a^2\rho$	$\frac{1}{2}a^2\rho$
total	$6a\rho$	$\bar{x}$	$\bar{y}$	$\frac{15}{2}a^2\rho$	$\frac{5}{2}a^2\rho$

ρ = mass per unit area $\quad x, y$ coords. taken horiz./ vert. from B — M2 A2

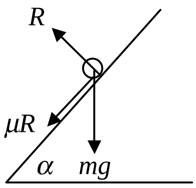
$\bar{x} = \dfrac{\frac{15}{2}a^2\rho}{6a\rho} = \frac{5a}{4}$ from AB — M1 A1

$\bar{y} = \dfrac{\frac{5}{2}a^2\rho}{6a\rho} = \frac{5a}{12}$ from BC — M1 A1

(b) $2a - \frac{5a}{12} = \frac{19a}{12}$ — A1



$\tan\theta = \dfrac{\frac{5}{4}a}{\frac{19}{12}a} = \frac{15}{19} \;\therefore\; \theta = 38°$ (nearest degree) — M2 A1 **(12)**

5.	*(a)*	$\frac{P}{v} - R - mg\sin\alpha = 0$	M1 A1	
		$\frac{P}{20} - 4400 - 40000(9.8)\frac{1}{20} = 0$	M1	
		$P = 20(4400 + 19600) = 480\,000$ W $= 480$ kW	M1 A1	
	(b)	$\frac{P}{v} - R = ma \;\; \therefore \;\; \frac{480000}{20} - 4400 = 40000a$	M1 A1	
		$a = 0.49$ ms^{-2}	A1	
	(c)	at max. speed, $a = 0 \;\; \therefore \;\; \frac{P}{v} - R = 0$	M1	
		$\frac{480000}{v} - 4400 = 0$ so $v = 109$ ms^{-1} (3sf)	M1 A1	
	(d)	model not suitable – lorry unable to attain 109 ms^{-1} ($\approx$ 245 mph)	B2	**(13)**
6.	*(a)*	cons. of mom: $2M(U) + 0 = 2M(V) + 5M(4)$	M1	
		$U = V + 10$	A1	
		$\frac{4-V}{U-0} = \frac{3}{4} \;\; \therefore \; 4 - V = \frac{3}{4}U$	M1 A1	
		solve simul. giving $U = 8$	M1 A1	
	(b)	$s_y = {}^{-}\frac{1}{2}gt^2 = {}^{-}19.6, \;\; t^2 = 4 \;\; \therefore \; t = 2$	M2 A1	
	(c)	$v_x = 4, \;\; v_y = 0 - gt = {}^{-}19.6$	M1 A1	
		req'd angle $= \tan^{-1}\frac{19.6}{4} = 78.5°$ (3sf) below horizontal	M1 A1	**(13)**
7.	*(a)*			
		m = mass of P $\quad d = AB$		
		resolve perp. to plane: $R - mg\cos\alpha = 0 \;\; \therefore \; R = mg(\frac{3}{5})$	M1 A1	
		frictional force $= \mu R = \frac{12}{35}mg$	A1	
		work done against friction = loss in KE – gain in PE	M1	
		$\frac{12}{35}mgd = \frac{1}{2}m(5.6)^2 - mgd\sin\alpha = 15.68m - \frac{4}{5}mgd$	M2 A2	
		$\frac{40}{35}gd = \frac{1}{2}(5.6)^2 \;\; \therefore \; d = 1.4\ m$	M1 A1	
	(b)	work done against friction = loss in KE (as PE returns to initial value)		
		$\frac{12}{35}mg \times 2.8 = \frac{1}{2}m(5.6^2 - v^2)$	M2 A1	
		$1.92g = 5.6^2 - v^2$	M1	
		$v^2 = 12.544 \;\; \therefore \; v = 3.5$ ms^{-1} (2sf)	M1 A1	**(16)**
			Total	**(75)**

Performance Record – M2 Paper E

Question no.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	Total
Topic(s)	**i**, **j** impulse	variable accel.	statics	centre of mass	power	collisions, projectiles	work - energy	
Marks	4	6	11	12	13	13	16	75
Student								

GCE Examinations

Mechanics Module M2

Advanced Subsidiary / Advanced Level

Paper E

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use $g = 9.8 \text{ m s}^{-2}$.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. A ball of mass 0.6 kg bounces against a wall and is given an impulse of (12**i** – 9**j**) N s where **i** and **j** are perpendicular horizontal unit vectors. The velocity of the particle after the impact is (5**i** + 3**j**) m s^{-1}.

Find the velocity of the particle before the impact. **(4 marks)**

---

2. A particle P moves along the x-axis such that its displacement, x metres, from the origin O at time t seconds is given by

$$x = 2 + t - \tfrac{1}{10}\text{e}^t.$$

(a) Find the distance of P from O when $t = 0$. **(2 marks)**

(b) Find, correct to 1 decimal place, the value of t when the velocity of P is zero. **(4 marks)**

3.

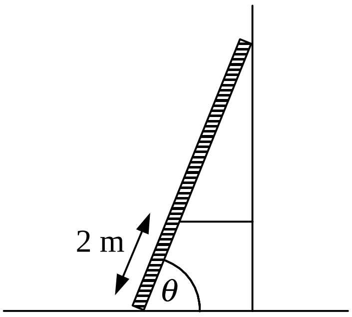


Fig. 1

Figure 1 shows a ladder of mass 20 kg and length 6 m leaning against a rough vertical wall with its lower end on smooth horizontal ground. The ladder is prevented from slipping along the ground by a light rope which is attached to the ladder 2 m from its bottom end and fastened to the wall so that the rope is horizontal and perpendicular to the wall.

The ladder is at an angle θ to the horizontal where $\tan\theta = \frac{5}{2}$ and the coefficient of friction between the ladder and the wall is $\frac{1}{3}$.

(a) Draw a diagram showing all the forces acting on the ladder. **(2 marks)**

(b) Show that the magnitude of the tension in the rope is $5g$. **(7 marks)**

A man wishes to use the ladder but fears the rope will snap as he climbs the ladder.

(c) Suggest, giving a reason for your answer, a more suitable position for the rope.

(2 marks)

4.

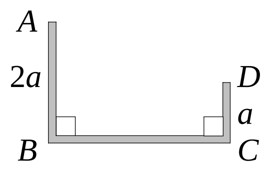


Fig. 2

Figure 2 shows an earring consisting of a uniform wire *ABCD* of length $6a$ bent to form right angles at *B* and *C* such that *AB* and *CD* are of length $2a$ and a respectively.

(a) Find, in terms of a, the distance of the centre of mass from

(i) *AB*,

(ii) *BC*. **(8 marks)**

The earring is to be worn such that it hangs in equilibrium suspended from the point *A*.

(b) Find, to the nearest degree, the angle made by *AB* with the downward vertical. **(4 marks)**

---

5. A lorry of mass 40 tonnes moves up a straight road inclined at an angle α to the horizontal where $\sin \alpha = \frac{1}{20}$. The lorry moves at a constant speed of 20 m s^{-1}.

In a model of the motion of the lorry, the non-gravitational resistance to motion is assumed to be constant and of magnitude 4400 N.

(a) Show that the engine of the lorry is working at a rate of 480 kW. **(5 marks)**

The road becomes horizontal. The lorry's engine continues to work at the same rate and the resistance to motion is assumed to remain unchanged.

(b) Find the initial acceleration of the lorry. **(3 marks)**

(c) Find, correct to 3 significant figures, the maximum speed of the lorry. **(3 marks)**

(d) Using your answer to part *(c)*, comment on the suitability of the modelling assumption.

(2 marks)

Turn over

6. Particle S of mass $2M$ is moving with speed U m s^{-1} on a smooth horizontal plane when it collides directly with a particle T of mass $5M$ which is lying at rest on the plane. The coefficient of restitution between S and T is $\frac{3}{4}$.

Given that the speed of T after the collision is 4 m s^{-1},

(a) find U. **(6 marks)**

As a result of the collision, T is projected horizontally from the top of a building of height 19.6 m and falls freely under gravity. T strikes the ground at the point X as shown in Figure 3.

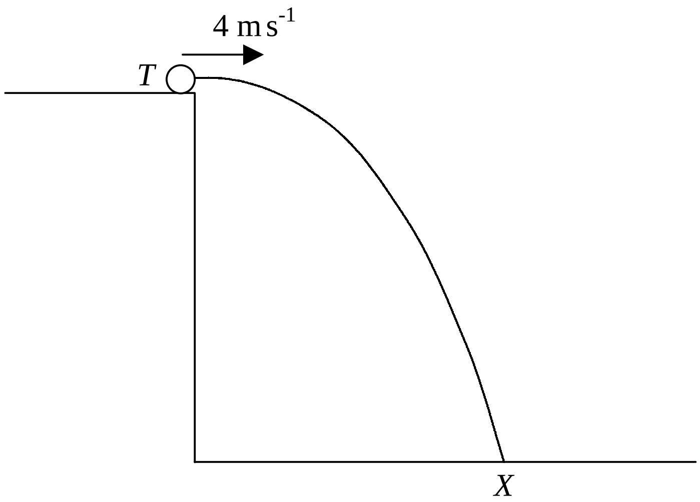


Fig. 3

(b) Find the time taken for T to reach X. **(3 marks)**

(c) Show that the angle between the horizontal and the direction of motion of T, just before it strikes the ground at X, is 78.5° correct to 3 significant figures. **(4 marks)**

7.

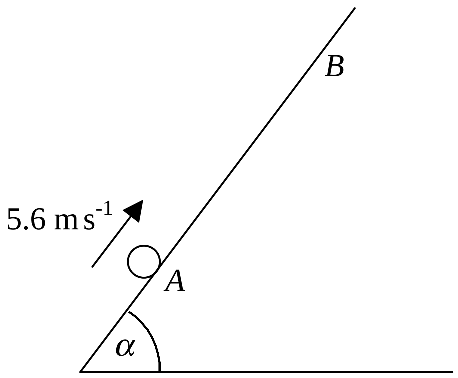


Fig. 4

Figure 4 shows a particle P projected from the point A up the line of greatest slope of a rough plane which is inclined at an angle α to the horizontal where $\sin \alpha = \frac{4}{5}$. P is projected with speed 5.6 m s^{-1} and the coefficient of friction between P and the plane is $\frac{4}{7}$.

Given that P first comes to rest at point B,

(a) use the Work-Energy principle to show that the distance AB is 1.4 m. **(10 marks)**

The particle then slides back down the plane.

(b) Find, correct to 2 significant figures, the speed of P when it returns to A. **(6 marks)**

END

GCE Examinations

Advanced Subsidiary / Advanced Level

Mechanics
Module M2

Paper F

MARKING GUIDE

This guide is intended to be as helpful as possible to teachers by providing concise solutions and indicating how marks should be awarded. There are obviously alternative methods that would also gain full marks.

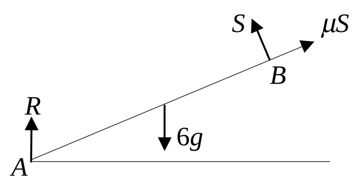
Method marks (M) are awarded for knowing and using a method.

Accuracy marks (A) can only be awarded when a correct method has been used.

(B) marks are independent of method marks.

Written by Shaun Armstrong & Chris Huffer

M2 Paper F – Marking Guide

1.		$\mathbf{I} = \Delta\text{mom.} = 0.5[(13\mathbf{i} + 7\mathbf{j}) - (5\mathbf{i} - 8\mathbf{j})]$	M1 A1	
		$= 0.5(8\mathbf{i} + 15\mathbf{j})$	A1	
		mag. of $\mathbf{I} = 0.5\sqrt{(8^2 + 15^2)} = 8.5$ Ns	M1 A1	**(5)**
2.	*(a)*	change in KE $= \frac{1}{2}\,1000(10^2 - 20^2) = {}^-150\,000$ J	M1 A1	
		change in PE $= 1000(9.8)(200\sin\theta) = 280\,000$ J	M2 A1	
		change in ME $= 280\,000 - 150\,000 =$ increase of $130\,000$ J	A1	
	(b)	air resistance	B1	
		friction	B1	**(8)**
3.	*(a)*	$s = t(2t^2 - 13t + 20) = t(2t - 5)(t - 4)$	M1 A1	
		particle at O when $s = 0$ $\therefore$ at $t = 0,\ \frac{5}{2},\ 4$ seconds	M1 A1	
	(b)	at rest when $v = 0$, $v = \frac{\mathrm{d}s}{\mathrm{d}t} = 6t^2 - 26t + 20$	M1 A1	
		$\therefore 3t^2 - 13t + 10 = 0$, $(t - 1)(3t - 10) = 0$	M1	
		$t = 1,\ \frac{10}{3}$ seconds	A1	**(8)**
4.	*(a)*			
		mom. about B: $6g\cos 30° - R.2\cos 30° = 0$	M1 A1	
		$\therefore R = 3g$	A1	
		mom. about A: $6g\cos 30° - S.2 = 0$	M1 A1	
		$\therefore S = \frac{3}{2}\sqrt{3}g$	A1	
	(b)	resolve $\rightarrow$: $\mu S\sin 60° - S\sin 30° = 0$	M1 A1	
		$\mu = \dfrac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \dfrac{1}{\sqrt{3}}$	A1	**(9)**
5.	*(a)*	at max. ht., $v_y = 0$ $\therefore 0 = (22\sin\alpha)^2 - 2gs$	M1 A1	
		$s_y = \dfrac{(22.\frac{7}{8})^2}{2g} = 18.91$	M1	
		starts 1.6 m above P so max. ht. above ground = 20.5 m (3sf)	A1	
	(b)	$s_y = {}^-1.4$ $\therefore ut\sin\alpha - \frac{1}{2}gt^2 = {}^-1.4$		
		$\frac{77}{4}t - 4.9t^2 = {}^-1.4$	M1 A1	
		$14t^2 - 55t - 4 = 0$ $\therefore (14t + 1)(t - 4) = 0$	M1	
		$t = 4$ in this case $\therefore$ ball in flight for 4 seconds	A1	
	(c)	$s_x = ut\cos\alpha = 22 \times 4 \times \frac{\sqrt{15}}{8} = 11\sqrt{15} = 42.6$	M1 A1	
		max. dist. fielder can run is $4 \times 6 = 24$ m	A1	
		max. initial dist. between fielder and ball $= 42.6 + 24 = 66.6$ m (3sf)	A1	**(12)**

6. *(a)* $\frac{1}{2}a$, since masses on AD are equal to mass at B — A1

(b)

portion	mass	y	my
lamina	$8m$	a	$8ma$
particle at A	$2m$	0	0
particle at B	$6m$	0	0
particle at D	$4m$	$2a$	$8ma$
total	$20m$	$\bar{y}$	$16ma$

y coords. taken vert. from AB — M2 A1

$\bar{y} = \frac{16ma}{20m} = \frac{4}{5}a$ — M1 A1

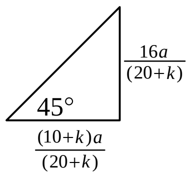
(c)

portion	mass	x	mx
lamina	$8m$	$\frac{a}{2}$	$4ma$
particle at A	$2m$	0	0
particles at B	$(6 + k)m$	a	$(6 + k)ma$
particle at D	$4m$	0	0
total	$(20 + k)m$	$\bar{x}$	$(10 + k)\,ma$

x coords. taken horiz. from AD — M1 A1

$\bar{x} = \frac{(10+k)ma}{(20+k)m} = \frac{(10+k)a}{(20+k)}$ — M1 A1

(d) new $\bar{y} = \frac{16ma}{(20+k)m} = \frac{16a}{(20+k)}$ — M2 A1



$\tan 45° = \frac{16a}{(10+k)a}$ $\therefore$ $1 = \frac{16}{10+k}$ giving $k = 6$ — M2 A1 **(16)**

---

7. *(a)* cons. of mom: $7u_1 + 4u_2 = 7(\frac{u_1}{2}) + 4v_2$ — M1

$8v_2 = 7u_1 + 8u_2$ — A1

$\frac{v_2 - \frac{1}{2}u_1}{u_1 - u_2} = e$ $\therefore$ $v_2 = eu_1 - eu_2 + \frac{1}{2}u_1$ — M1 A1

eliminate v_2 giving $7u_1 + 8u_2 = 8eu_1 - 8eu_2 + 4u_1$ — M1 A1

$8u_2 + 8eu_2 = 8eu_1 - 3u_1$ $\therefore$ $8u_2(e + 1) = u_1(8e - 3)$ — A1

(b) sub. in for u_1 and u_2: $24(e + 1) = 14(8e - 3)$ — M1

$24e + 24 = 112e - 42$ giving $e = \frac{3}{4}$ — M1 A1

(c) speeds of A, B after impact are v_1 and v_2 resp.

$v_1 = 7\text{ ms}^{-1}$, $v_2 = (\frac{7}{8})14 + 3 = 15.25\text{ ms}^{-1}$ — A1

original KE $= \frac{1}{2} \times 7 \times 14^2 + \frac{1}{2} \times 4 \times 3^2 = 704$ J — M1 A1

final KE $= \frac{1}{2} \times 7 \times 7^2 + \frac{1}{2} \times 4 \times 15.25^2 = 636.625$ J — M1 A1

% KE lost $= \frac{704 - 636.625}{704} \times 100 = 9.6\%$ (2sf) — M1 A1 **(17)**

Total **(75)**

Performance Record – M2 Paper F

Question no.	1	2	3	4	5	6	7	Total
Topic(s)	**i**, **j** impulse	energy	variable accel.	statics	projectiles	centre of mass	collisions, energy	
Marks	5	8	8	9	12	16	17	75
Student								

GCE Examinations

Mechanics
Module M2

Advanced Subsidiary / Advanced Level

Paper F

Time: 1 hour 30 minutes

Instructions and Information

Candidates may use any calculator except those with a facility for symbolic algebra and / or calculus.

Full marks may be obtained for answers to ALL questions.

Mathematical and statistical formulae and tables are available.

This paper has 7 questions.

When a numerical value of g is required, use $g = 9.8 \text{ m s}^{-2}$.

Advice to Candidates

You must show sufficient working to make your methods clear to an examiner. Answers without working will gain no credit.

Written by Shaun Armstrong & Chris Huffer

1. An ice hockey puck of mass 0.5 kg is moving with velocity $(5\mathbf{i} - 8\mathbf{j})$ m s^{-1}, where $\mathbf{i}$ and $\mathbf{j}$ are perpendicular horizontal unit vectors, when it is struck by a stick. After the impact, the puck travels with velocity $(13\mathbf{i} + 7\mathbf{j})$ m s^{-1}.

Find the magnitude of the impulse exerted by the stick on the puck. **(5 marks)**

---

2. A car of mass 1 tonne is climbing a hill inclined at an angle θ to the horizontal where $\sin\theta = \frac{1}{7}$. When the car passes a point X on the hill, it is travelling at 20 m s^{-1}. When the car passes the point Y, 200 m further up the hill, it has speed 10 m s^{-1}.

In a preliminary model of the situation, the car engine is assumed only to be doing work against gravity. Using this model,

(a) find the change in the total mechanical energy of the car as it moves from X to Y.

(6 marks)

In a more sophisticated model, the car engine is also assumed to work against other forces.

(b) Write down two other forces which this model might include. **(2 marks)**

---

3. A particle moves along a straight horizontal track such that its displacement, s metres, from a fixed point O on the line after t seconds is given by

$$s = 2t^3 - 13t^2 + 20t.$$

(a) Find the values of t for which the particle is at O. **(4 marks)**

(b) Find the values of t at which the particle comes instantaneously to rest. **(4 marks)**

Turn over

4.

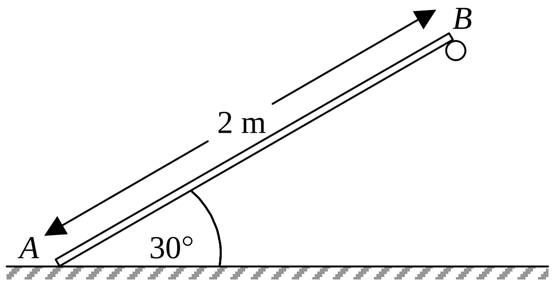


Fig. 1

Figure 1 shows a uniform rod *AB* of length 2 m and mass 6 kg inclined at an angle of 30° to the horizontal with *A* on smooth horizontal ground and *B* supported by a rough peg. The rod is in limiting equilibrium and the coefficient of friction between *B* and the peg is μ.

(a) Find, in terms of *g*, the magnitude of the reactions at *A* and *B*. **(6 marks)**

(b) Show that $\mu = \frac{1}{\sqrt{3}}$. **(3 marks)**

5.

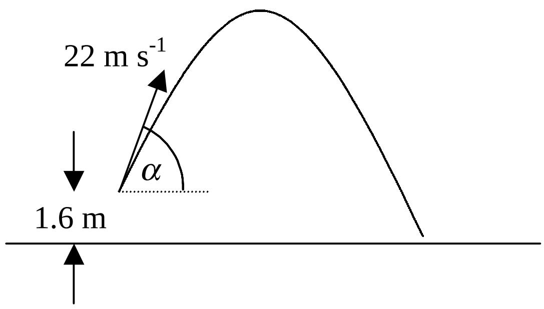


Fig. 2

During a cricket match, a batsman hits the ball giving it an initial velocity of 22 m s^{-1} at an angle α to the horizontal where $\sin\alpha = \frac{7}{8}$. When the batsman strikes the ball it is 1.6 metres above the ground, as shown in Figure 2, and it subsequently moves freely under gravity.

(a) Find, correct to 3 significant figures, the maximum height above the ground reached by the ball.
(4 marks)

The ball is caught by a fielder when it is 0.2 metres above the ground.

(b) Find the length of time for which the ball is in the air. **(4 marks)**

Assuming that the fielder who caught the ball ran at a constant speed of 6 m s^{-1},

(c) find, correct to 3 significant figures, the maximum distance that the fielder could have been from the ball when it was struck.
(4 marks)

Turn over

6.

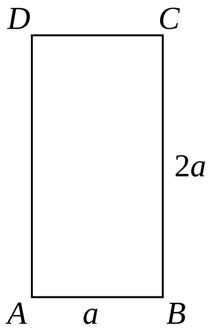


Fig. 3

Figure 3 shows a uniform rectangular lamina *ABCD* of mass $8m$ in which the sides *AB* and *BC* are of length a and $2a$ respectively. Particles of mass $2m$, $6m$ and $4m$ are fixed to the lamina at the points *A*, *B* and *D* respectively.

(a) Write down the distance of the centre of mass from *AD*. **(1 mark)**

(b) Show that the distance of the centre of mass from *AB* is $\frac{4}{5}a$. **(5 marks)**

Another particle of mass km is attached to the lamina at the point *B*.

(c) Show that the distance of the centre of mass from *AD* is now given by $\frac{(10+k)a}{20+k}$.

(4 marks)

Given that when the lamina is suspended freely from the point *A* the side *AB* makes an angle of 45° with the vertical,

(d) find the value of k. **(6 marks)**

---

7. Particle *A* of mass 7 kg is moving with speed u_1 on a smooth horizontal surface when it collides directly with particle *B* of mass 4 kg moving in the same direction as *A* with speed u_2.

After the impact, *A* continues to move in the same direction but its speed has been halved. Given that the coefficient of restitution between the particles is e,

(a) show that $8u_2(e + 1) = u_1(8e - 3)$. **(7 marks)**

Given also that $u_1 = 14 \text{ m s}^{-1}$ and $u_2 = 3 \text{ m s}^{-1}$,

(b) find e, **(3 marks)**

(c) show that the percentage of the kinetic energy of the particles lost as a result of the impact is 9.6%, correct to 2 significant figures.

(7 marks)

END